Data Structure Using C Lab (24M15CA111)

ODD 2025

1. Write a program to input and display elements of an array.
2. Find the sum and average of elements of an array.
3. Find the maximum and minimum element in an array.
4. Reverse the elements of an array.
5. Count the frequency of each element in an array.
6. Implement linear search on an array.
7. Sort the array using bubble sort.
8. Sort the array using selection sort.
9. Sort the array using insertion sort.
10. Merge two sorted arrays into one sorted array.
11. Rotate an array by k positions (left and right).
12. Find the second largest and second smallest element in an array.
13. Rearrange the array so that all even numbers appear before odd numbers.
14. Read and display a matrix of size $m \times n$.
15. Find the sum of all elements of a matrix.
16. Add two matrices.
17. Multiply two matrices.
18. Find the transpose of a matrix.
19. Calculate the sum of the principal and secondary diagonals of a square matrix.
20. Check if a square matrix is symmetric.
21. Find the determinant of a 2×2 or 3×3 matrix.
22. Find the maximum element in each row and each column of a matrix.
23. Find the row with the maximum sum and the column with the maximum sum.